

An Analysis of RMSprop optimizer

RMSprop optimizer

RMSprop optimizer -- Part 1

This equation is implicit since w_{new} appears on both sides of the equation. It is a stochastic form of the proximal gradient method since the update can also be written as:

RMSprop optimizer -- Part 2

$$\begin{bmatrix} w_1 \\ w_2 \end{bmatrix} \leftarrow \begin{bmatrix} w_1 \\ w_2 \end{bmatrix} - \eta \begin{bmatrix} \partial \mathcal{L}(w_1 + w_2 x_i - y_i) \\ \partial \mathcal{L}(w_1 + w_2 x_i - y_i) \end{bmatrix} = \begin{bmatrix} w_1 \\ w_2 \end{bmatrix} - \eta \begin{bmatrix} 2(w_1 + w_2 x_i - y_i) x_i \\ 2(w_1 + w_2 x_i - y_i) \end{bmatrix}.$$
$$\begin{bmatrix} w_1 \\ w_2 \end{bmatrix} \leftarrow \begin{bmatrix} w_1 \\ w_2 \end{bmatrix} - \eta \begin{bmatrix} \frac{\partial \mathcal{L}(w_1 + w_2 x_i - y_i)}{\partial w_1} \\ \frac{\partial \mathcal{L}(w_1 + w_2 x_i - y_i)}{\partial w_2} \end{bmatrix}$$
$$x_i = \begin{bmatrix} w_1 \\ w_2 \end{bmatrix} - \eta \begin{bmatrix} 2(w_1 + w_2 x_i - y_i) x_i \\ 2(w_1 + w_2 x_i - y_i) \end{bmatrix}.$$

Note that in each iteration or update step, the gradient is only evaluated at a single x_i . This is the key difference between stochastic gradient descent and batched gradient descent. In general, given a linear regression $\hat{y} = \sum_{k=1}^m w_k x_k$ problem, stochastic gradient descent behaves differently when $m < n$ (underparameterized) and $m \geq n$ (overparameterized). In the overparameterized case, stochastic gradient descent converges to $\arg \min_{w: w^T x_k = y_k \forall k \in 1:n} \|w - w_0\|$. That is, SGD converges to the interpolation solution with minimum distance from the starting w_0 . This is true even when the learning rate remains constant. In the underparameterized case, SGD does not converge if learning rate remains constant.

RMSprop optimizer -- Part 3

$G_j, j = 1, 2, \dots, \tau$. This vector essentially stores a historical sum of gradient squares by dimension and is updated after every iteration. The formula for an update is now

RMSprop optimizer -- Part 4

Sign-based stochastic gradient descent Even though sign-based optimization goes back to the aforementioned Rprop, in 2018 researchers tried to simplify Adam by removing the magnitude of the stochastic gradient from being taken into account and only considering its sign. This results in a significantly lower

communication cost of transferring gradients from workers to the parameter server. In this sense, it serves to better compress the gradient information, while having comparable convergence to standard SGD.

RMSprop optimizer -- Part 5

Implicit updates (ISGD) As mentioned earlier, classical stochastic gradient descent is generally sensitive to learning rate η . Fast convergence requires large learning rates but this may induce numerical instability. The problem can be largely solved by considering implicit updates whereby the stochastic gradient is evaluated at the next iterate rather than the current one:

RMSprop optimizer -- Part 6

for $\Sigma(w) = \frac{1}{n} \sum_{i=1}^n Q_i(w) - Q(w)$ where dB_t denotes the Ito-integral with respect to a Brownian motion is a more precise approximation in the sense that there exists a constant $C > 0$ such that

RMSprop optimizer -- Part 7

Adam Adam (short for Adaptive Moment Estimation) is a 2014 update to the RMSprop optimizer combining it with the main feature of the Momentum method. In this optimization algorithm, running averages with exponential forgetting of both the gradients and the second moments of the gradients are used. Given parameters $w(t)$ and a loss function $L(t)$, where t indexes the current training iteration (indexed at 1), Adam's parameter update is given by:

RMSprop optimizer -- Part 8

where $E\{\cdot\}$ denotes taking the expectation with respect to the random choice of indices in the stochastic gradient descent scheme. Since this approximation does not capture the random fluctuations around the mean behavior of stochastic gradient descent solutions to stochastic differential equations (SDEs) have been proposed as limiting objects. More precisely, the solution to the SDE

RMSprop optimizer -- Part 9

The step size is denoted by η (sometimes called the learning rate in machine learning) and here $:=$ denotes the update of a variable in the

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algorithm. In many cases, the summand functions have a simple form that enables inexpensive evaluations of the sum-function and the sum gradient. For example, in statistics, one-parameter exponential families allow economical function-evaluations and gradient-evaluations. However, in other cases, evaluating the sum-gradient may require expensive evaluations of the gradients from all summand functions. When the training set is enormous and no simple formulas exist, evaluating the sums of gradients becomes very expensive, because evaluating the gradient requires evaluating all the summand functions' gradients. To economize on the computational cost at every iteration, stochastic gradient descent samples a subset of summand functions at every step. This is very effective in the case of large-scale machine learning problems.

RMSprop optimizer -- Part 10

RMSProp has shown good adaptation of learning rate in different applications. RMSProp can be seen as a generalization of Rprop and is capable to work with mini-batches as well opposed to only full-batches.

RMSprop optimizer -- Part 11

Second-order methods A stochastic analogue of the standard (deterministic) Newton–Raphson algorithm (a "second-order" method) provides an asymptotically optimal or near-optimal form of iterative optimization in the setting of stochastic approximation. A method that uses direct measurements of the Hessian matrices of the summands in the empirical risk function was developed by Byrd, Hansen, Nocedal, and Singer. However, directly determining the required Hessian matrices for optimization may not be possible in practice. Practical and theoretically sound methods for second-order versions of SGD that do not require direct Hessian information are given by Spall and others. (A less efficient method based on finite differences, instead of simultaneous perturbations, is given by Ruppert.) Another approach to the approximation Hessian matrix is replacing it with the Fisher information matrix, which transforms usual gradient to natural. These methods not requiring direct Hessian information are based on either values of the summands in the above empirical risk function or values of the gradients of the summands (i.e., the SGD inputs). In particular, second-order optimality is asymptotically achievable without direct calculation of the Hessian matrices of the summands in the empirical risk function. When the objective is a nonlinear least-squares loss